



PIZZA



SALES ANALYSIS

ANALYZING TRENDS, SALES PERFORMANCE & CUSTOMER
PREFERENCES USING SQL



MADHAVI PATHAK | DATA ANALYTICS





UNDERSTANDING PIZZA SALES TRENDS TO OPTIMIZE BUSINESS DECISIONS.

TOOLS USED: MYSQL, EXCEL

- Total Orders: 48,000+ pizza orders
- Time Frame: January 2015 – December 2015 (1 year)
- Number of Pizzas Sold: 91,000+
- Total Revenue: \$1.7M+
- Pizza Categories: Classic, Veggie, Chicken, Supreme, etc.
- Columns in the Dataset: Order ID, Date, Pizza Name, Category, Size, Price, Quantity, Total Sales



WHAT ARE WE ANALYZING?

Monthly & yearly sales trends 📊

Best-selling pizza categories 🍕

Sales distribution by region 🌍

Revenue trends & customer preferences 🏆

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.



```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

	total_orders
▶	21350



CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    ROUND(SUM(o.quantity * p.price), 2) AS Tot_Revenue
FROM
    order_details AS o
    JOIN
    pizzas AS p ON p.pizza_id = o.pizza_id;
```

	Tot_Revenue
▶	817860.05

IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

	name	price
▶	The Greek Pizza	35.95

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    quantity, COUNT(order_details_id)
FROM
    order_details
GROUP BY quantity;
```

	quantity	COUNT(order_details_id)
▶	1	47693
	2	903
	3	21
	4	3

```
SELECT
    pizzas.size, COUNT(order_details_id)
FROM
    order_details
    JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id
GROUP BY size
ORDER BY COUNT(order_details_id) DESC;
```

	size	COUNT(order_details_id)
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pizza_types.name, SUM(quantity) AS total_orders
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY name
ORDER BY total_orders DESC
LIMIT 5;
```

	name	total_orders
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY.

```
SELECT
    pizza_types.category, SUM(quantity) AS total_orders
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY category
ORDER BY total_orders DESC;
```

category	total_orders
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    HOUR(orders_time), COUNT(order_id) AS total_orders
FROM
    orders
GROUP BY HOUR(orders_time)
ORDER BY total_orders DESC;
```

Result Grid			Filter Rows:
	HOUR(orders_time)	total_orders	
▶	12	2520	
	13	2455	
	18	2399	
	17	2336	
	19	2009	
	16	1920	
	20	1642	
	14	1472	
	15	1468	
	11	1231	
	21	1198	
	22	663	
	23	28	
	10	8	
	9	1	

JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT  
    category, COUNT(pizza_type_id)  
FROM  
    pizza_types  
GROUP BY category;
```

	category	count(pizza_type_id)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```
SELECT
    ROUND(AVG(tot_quantity), 0) AS avg_pizza_orders_perday
FROM
    (SELECT
        orders_date, SUM(quantity) AS tot_quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders_date) AS order_quantity;
```

	avg_pizza_orders_perday
▶	138

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
select pizza_types.category, round((sum(order_details.quantity * pizzas.price)/(SELECT  
ROUND(SUM(o.quantity * p.price), 2) AS Tot_Revenue  
FROM order_details AS o JOIN  
pizzas AS p ON p.pizza_id = o.pizza_id) ) * 100, 2) as revenue  
from pizza_types join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join order_details on pizzas.pizza_id = order_details.pizza_id  
group by pizza_types.category order by revenue desc;
```

	category	revenue
▶	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.



```
select orders_date, sum(revenue) over(order by orders_date) as cum_revenue
from (select orders.orders_date, sum(order_details.quantity * pizzas.price)
as revenue
from order_details join pizzas on pizzas.pizza_id = order_details.pizza_id
join orders on order_details.order_id = orders.order_id
group by orders.orders_date) as sales;
```

	orders_date	cum_revenue
▶	2015-01-01	2713.85000000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.3500000000002
	2015-01-11	25862.65



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
select category, name, revenue, ranking
from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as ranking
from
(select pizza_types.category, pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as sale) as top
where ranking<= 3;
```

	category	name	revenue	ranking
▶	Chicken	The Thai Chicken Pizza	43434.25	1
	Chicken	The Barbecue Chicken Pizza	42768	2
	Chicken	The California Chicken Pizza	41409.5	3
	Classic	The Classic Deluxe Pizza	38180.5	1
	Classic	The Hawaiian Pizza	32273.25	2
	Classic	The Pepperoni Pizza	30161.75	3
	Supreme	The Spicy Italian Pizza	34831.25	1
	Supreme	The Italian Supreme Pizza	33476.75	2
	Supreme	The Sicilian Pizza	30940.5	3
	Veggie	The Four Cheese Pizza	32265.70000000065	1
	Veggie	The Mexicana Pizza	26780.75	2
	Veggie	The Five Cheese Pizza	26066.5	3



THANK YOU

